

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250287631

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Pan; Kuan-Ting et al.

Gate Isolation Feature and Manufacturing Method Thereof

Abstract

A semiconductor structure includes a plurality of active regions extending along a first direction, a plurality of gate structure segments positioned along a line extending in a second direction, the second direction being orthogonal to the first direction, wherein the gate structure segments are separated by dummy structures. The semiconductor structure further includes a conductive layer disposed over both the gate structure segments and the dummy structures to electrically connect at least some of the gate structure segments, and a cut feature aligned with one of the dummy structures and positioned to electrically isolate gate structure segments on both sides of the one of the dummy structures.

Inventors: Pan; Kuan-Ting (Hsinchu, TW), Su; Huan-Chieh (Changhua County, TW), You; Jia-Chuan (Taoyuan County, TW), Ju; Shi Ning (Hsinchu City, TW), Chiang; Kuo-Cheng (Hsinchu County, TW), Jhan; Yi-Ruei (Hsinchu, TW), Chuang; Li-Yang (Hsinchu, TW), Wang; Chih-Hao (Hsinchu County, TW)

Applicant: Taiwan Semiconductor Manufacturing Co., Ltd. (Hsin-Chu, TW)

Family ID: 1000008619443

Appl. No.: 19/214597

Filed: May 21, 2025

Related U.S. Application Data

parent US continuation 18750379 20240621 parent-grant-document US 12317528 child US 19214597

parent US continuation 18361556 20230728 parent-grant-document US 12021136 child US 18750379

parent US division 17091767 20201106 parent-grant-document US 11799019 child US 18361556
us-provisional-application US 62982149 20200227

Publication Classification

Int. Cl.: **H10D30/01** (20250101); **H10D62/10** (20250101); **H10D84/01** (20250101); **H10D84/03** (20250101)

U.S. Cl.:

CPC **H10D30/0243** (20250101); **H10D62/115** (20250101); **H10D62/119** (20250101); **H10D84/0158** (20250101); **H10D84/038** (20250101);

Background/Summary

PRIORITY DATA [0001] The present application is a continuation application of U.S. patent application Ser. No. 18/750,379 filed Jun. 21, 2024, which is a continuation application of U.S. patent application Ser. No. 18/361,556, filed Jul. 28, 2023, now U.S. Pat. No. 12,021,136, which claims priority as a divisional application to U.S. patent application Ser. No. 17/091,767, filed Nov. 6, 2020, now U.S. Pat. No. 11,799,019, which claims the benefit of U.S. Patent Provisional Application No. 62/982,149, filed Feb. 27, 2020, each of which is hereby incorporated by reference in its entirety.

BACKGROUND

[0002] The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of IC structures (such as three-dimensional transistors) and processing and, for these advancements to be realized, similar developments in IC processing and manufacturing are needed. For example, device performance (such as device performance degradation associated with various defects) and fabrication cost of field-effect transistors become more challenging when device sizes continue to decrease. Although methods for addressing such a challenge have been generally adequate, they have not been entirely satisfactory in all aspects.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIGS. 1A, 1B, 1C, 1D, 1E, 1F, and 1G are diagrams showing an illustrative process for forming a gate cut feature, according to one example of principles described herein.

[0005] FIG. 2 is a diagram showing a top view of the gate cut feature, according to one example of principles described herein.

[0006] FIG. 3 is a flowchart showing an illustrative method for forming a gate cut feature, according to one example of principles described herein.

DETAILED DESCRIPTION

[0007] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0008] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0009] The present disclosure is generally related to semiconductor devices and the fabrication thereof, and more particularly to methods of fabricating field-effect transistors (FETs), particularly, as Gate All-Around (GAA) transistors or fin-like FETs (FinFETs). In GAA transistor devices, the gate surrounds the channel on all sides. For example, the gate entirely encompasses one or more nanostructures that are suspended between active source/drain regions. GAA devices are formed by depositing alternating layers of different semiconductor materials (e.g., silicon and silicon germanium). These alternating layers may then be patterned to form fin structures. Then, after various other structures are put in place, one of the materials, such as the silicon germanium can be removed, thus leaving nanowires or nanosheets in place. Then, the gate layers (high-k, workfunction, metal gate) may then be formed so as to completely surround each of the nanowires and nanosheets.

[0010] FinFET devices provide for improved device performance over planar transistors because in a finFET device, the gate surrounds three sides of the channel. Conventional methods for forming a finFET device involve forming a dummy gate over a set of fin structures running in parallel. Sidewall spacers are then formed on the sidewalls of the gate. After the sidewall spacers are formed, source/drain regions may be formed on the fin structures on both sides of the gate. After the source/drain regions are formed, and an interlayer Dielectric Layer (ILD) is formed over the source/drain regions, the dummy gate can be replaced with a real gate that includes a conductive material such as a metal material.

[0011] One challenge with semiconductor fabrication is designing patterns while taking the spacing requirement of cut gate structures. Cut gate structures are formed by etching away a portion of a deposited gate and filling the trench with a dielectric material, thereby “cutting” the metal gate feature so that gate segments on both sides of the cut feature are electrically isolated from each other.

[0012] According to principles described herein, the spacing constraints for cut features and alignment process is substantially improved by changing the depth at which cut features are able to be made. Specifically, dielectric dummy fin structures are formed between the functional fin structures. In some instances, the dielectric dummy fin structure is also called as a hybrid fin structure. Then, a gate layer is deposited over both the functional fin structures and the dummy fin structures. The gate layer is then etched back to expose top surfaces of the dummy fin structures. Thus, each dummy fin structure separates the gate layer into different gate segments. Then, a

conductive layer, such as Tungsten, is deposited over the gate layer and the dummy fin structures. The conductive layer electrically connects the gate layer segments. To cut the gate layer, a cut feature is formed over one of the dummy fin structure. Thus, the cut feature “cuts” the conductive layer over the dummy fin structure and thus electrically isolates two adjacent gate layer segments. [0013] Using the principles described herein, the overlay constraints for cut features are reduced because the cut feature may be placed anywhere over a dummy fin structure. Furthermore, the circuit can be designed by spacing fin structures closer together, even where gate features are to be cut.

[0014] FIGS. 1A illustrates a workpiece that includes a semiconductor substrate **102** and fin structures **106a**, **106b**, **106c**, **106d** separated by shallow trench isolation (STI) regions **104**. The semiconductor substrate **102** may be a silicon substrate. The semiconductor substrate may be part of a silicon wafer. Other semiconductor materials are contemplated. The substrate **102** may include an elementary (single element) semiconductor, such as silicon, germanium, and/or other suitable materials; a compound semiconductor, such as silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, indium antimonide, and/or other suitable materials; an alloy semiconductor, such as SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, GaInAsP, and/or other suitable materials. The substrate **102** may be a single-layer material having a uniform composition. Alternatively, the substrate **102** may include multiple material layers having similar or different compositions suitable for IC device manufacturing. In one example, the substrate **102** may be a silicon-on-insulator (SOI) substrate having a silicon layer formed on a silicon oxide layer. In another example, the substrate **102** may include a conductive layer, a semiconductor layer, a dielectric layer, other layers, or combinations thereof.

[0015] The fin structures **104** may include stacked channel structures, such as nanostructures including nanowires or nanosheets. Such structures are used in gate-all-around (GAA) transistor devices. In GAA transistor devices, the gate surrounds the channel on all sides. For example, the gate entirely encompasses one or more nanostructures that are suspended between active source/drain regions. The principles described herein may be applied to fin structures that have a gate on three sides, as well as fin structures that are processed to include nanostructures. The example described herein includes nanostructures. Thus, the term fin structures as used herein may include a nanowire or nanosheet stack formed from a fin structure.

[0016] To form the fin structures **106a**, **106b**, **106c**, **106d**, alternating layers **110**, **112** of differing semiconductor materials may be deposited onto the substrate **102**. For example, if the substrate is a silicon substrate, then alternating layers of silicon and silicon germanium (SiGe) may be deposited. Then, a hard mask layer **108** may be deposited on top of the alternating semiconductor layers. The hardmask layer **108** may include at least one of silicon oxide (SiO₂), silicon nitride (SiN), silicon carbide (SiC), silicon oxynitride (SiON), silicon oxycarbide (SiOCN), hafnium oxide (HfO₂), aluminum oxide (Al₂O₃), and zirconium oxide (ZrO₂). Other materials are contemplated. A photoresist material may then be used to pattern the hard mask **108** layer. The photoresist may then be exposed to a light source through a photomask. The photoresist may then be developed such that the portions of the photoresist remain while other portions are removed. The pattern within the developed photomask is then transferred to the hard mask layer **108**, which is then transferred to the substrate **102** and alternating semiconductor layers through an etching process. This forms the fin structures **106a**, **106b**, **106c**, **106d** as shown. The fin structures comprise elongated fin-like structures that run parallel to each other.

[0017] After, the fin structures **106a**, **106b**, **106c**, **106d** are formed, an STI layer **104** may be deposited, and then etched back to the desired height. The STI layer **104** separates the fin structures from each other. The STI layer **104** may be recessed to a point approximately halfway between a bottom and a top of the fin structure. However, in some examples, the STI layer **104** may be recessed to different heights along the height of the fin structures **106a**, **106b**, **106c**, **106d**.

[0018] After the STI layer **104** is formed, a cladding material **105** may be conformally deposited

over the fin structures. The cladding material **105** may be the same material as one of the alternating semiconductor layers, particularly, the semiconductor material that is to be removed. Thus, in the example where the alternating layers are silicon **112** and silicon germanium **110**, then the cladding material **105** may be silicon germanium.

[0019] FIG. **1B** illustrates the formation of dummy fin structures **114a**, **114b**, **114c** between the real fin structures **106a**, **106b**, **106c**, **106d**. The dummy fin structures may be formed by several processes. Specifically, a dielectric material **116** may be conformally deposited within the trenches between the real fin structures **106a**, **106b**, **106c**, **106d**. This dielectric material may be, for example, silicon nitride (SiN), silicon oxynitride (SiON), or silicon carbide (SiC). Other dielectric materials may be used. The dielectric material **116** may be deposited using an Atomic Layer Deposition (ALD) process.

[0020] After the conformal dielectric layer **116** is formed, an oxide layer **118** is deposited on the dielectric layer **116**. The oxide layer **118** may be, for example, silicon oxide. In some examples, the oxide layer **118** may be the same type of material used for the STI layer **104**. The oxide layer **118** may be formed, for example, using a Chemical Vapor Deposition (CVD) process. Other processes may be used as well. After the oxide layer **118** is formed, a Chemical Mechanical Polishing (CMP) process may be applied to planarize the top surface of the workpiece. A CMP process involves applying a slurry to the surface of the workpiece. The slurry includes etching chemicals as well as solid particles. A polishing head is then moved across the surface of the workpiece and the chemical and mechanical forces on the workpiece result in removing material from the workpiece at a substantially similar rate so as to create a planar surface.

[0021] An etching process may then be applied to selectively remove the oxide layer **118** while leaving the cladding layer **105** substantially intact. The etching process may be, for example, a dry etching process. In some examples, the etching process is applied so that, after the etching process, the top surface of the oxide layer **118** is about 5-15 nanometers higher than a top surface of the top channel **112** of the fin structures **106a**, **106b**, **106c**, **106d**. If the height difference is smaller than 5 nanometers, a subsequent high-k dielectric layer increases, thereby increasing parasitic capacitance, in some instances. If the height difference is greater than 15 nanometers, a subsequent high-k material layer is insufficient to protect gate spacers during an etching operation for contact features, in some instances.

[0022] After the oxide layer **118** has been partially etched back, a high-k dielectric layer **120** may be deposited at the top of the dummy fin structures **114a**, **114b**, **114c**. The high-k dielectric layer **120** may fill the space left by the etching process that etches back the oxide layer **118**. The high-k dielectric layer may be, for example, hafnium oxide (HfO₂), zirconium oxide (ZrO₂), hafnium aluminum oxide (HfAlO_x), hafnium silicon oxide (HfSiO_x), or aluminum oxide (Al₂O₃). The bottom surface of the high-k dielectric layer **120** may be about 5-15 nanometers higher than a top surface of the top channel **112** of the fin structures **106a**, **106b**, **106c**, **106d**.

[0023] FIG. **1C** illustrates the formation of a temporary dummy gate **122**. The dummy gate is placed where a metal gate may ultimately be formed. Sidewall spacers (not shown) may be formed on both sides of the dummy gate. Then, source/drain regions (not shown) may be formed within the channel regions **112** of the fin structures **106a**, **106b**, **106c**, **106d**. The source/drain regions may be formed by removing part of the fin structure and replacing it with an epitaxially grown doped region. After the source/drain regions are formed, the dummy gate **122** may be removed.

[0024] FIG. **1D** illustrates removal of the cladding material **105** and the non-channel regions **110** of the fin structure **106a**, **106b**, **106c**, **106d**. In the example where the channel portions **112** are silicon and the non-channel regions **110** are silicon germanium, then the etching process used to remove the non-channel regions **110** and the cladding material **105** may be configured to selectively remove silicon germanium while leaving silicon substantially intact. In some examples, if the cladding material **105** is different than the non-channel regions **110**, then two separate etching

processes may be used to remove the cladding material **105** and the non-channel regions **110**. In either case, an etching process used to remove such regions may be, for example, a wet etching process. With the channel regions **112** exposed, the process for forming the real metal gate may be started. This may involve forming various layers around the channel regions **112**, such as high-k gate layers (not shown) and/or workfunction layers (not shown). Such workfunction metal is designed to give metal gates the desired properties for ideal functionality. Various examples of a p-type workfunction metal may include, but are not limited to, tungsten carbon nitride (WCN), tantalum nitride (Ta₂N), titanium nitride (TiN), titanium aluminum nitride (TiAlN), tungsten sulfur nitride (TSN), tungsten (W), cobalt (Co), molybdenum (Mo), etc. Various examples of n-type workfunction metals include, but are not limited to, aluminum (Al), titanium aluminum (TiAl), titanium aluminum carbide (TiAlC), titanium aluminum silicon carbide (TiAlSiC), tantalum aluminum silicon carbide (TaAlSiC), and hafnium carbide (HfC).

[0025] In some examples, the wet etching process may use an acid-based etchant such as: sulfuric acid (H₂SO₄), perchloric acid (HClO₄), hydroiodic acid (HI), hydrobromic acid (HBr), nitric acid (HNO₃), hydrochloric acid (HCl), acetic acid (CH₃COOH), citric acid (C₆H₈O₇), potassium periodate (KIO₄), tartaric acid (C₄H₆O₆), benzoic acid (C₆H₅COOH), tetrafluoroboric acid (HBF₄), carbonic acid (H₂CO₃), hydrogen cyanide (HCN), nitrous acid (HNO₂), hydrofluoric acid (HF), or phosphoric acid (H₃PO₄). In some examples, an alkaline-based etchant may be used. Such etchants may include but are not limited to ammonium hydroxide (NH₄OH) and potassium hydroxide (KOH).

[0026] FIG. 1E illustrates formation of metal gate segments **124a**, **124b**, **124c**, **124d** for each of the fin structures **106a**, **106b**, **106c**, **106d**. Formation of the metal gate segments may begin by depositing a metal material **124** within the spaces left by the previous removal process. The metal material **124** surrounds each of the channel regions **112**. After the metal material **124** has been deposited, the metal material **124** may be etched back so that the top surface of the metal material **124** is lower than a top surface of the dummy fin structures **114a**, **114b**, **114c**. This etching process may be, for example, a dry etching process. The etching process exposes the top surface of the dummy fin structures **114a**, **114b**, **114c**. The etching process may be applied such that the top surface of the metal gate segments **124a**, **124b**, **124c**, **124d** is less than 2 nanometers from the top surface of the dummy fin structures **114a**, **114b**, **114c**. If the height difference is greater than 2 nanometers, a portion of work function metal(s) is damaged, which impacts a device threshold voltage, in some instances.

[0027] FIG. 1F illustrates the formation of a conductive layer **126** that covers both the metal gate segments **124a**, **124b**, **124c**, **124d** and the top surfaces of the dummy fin structures **114a**, **114b**, **114c**. The conductive layer **126** thus electrically connects the gate segments **124a**, **124b**, **124c**, **124d** to each other. The conductive layer **126** includes at least one of tungsten (W), cobalt (Co), ruthenium (Ru), and copper (Cu). In some examples, a thin layer is optionally deposited before the conductive layer is deposited. The thin layer includes at least one of titanium (Ti), titanium nitride (TiN), tantalum (Ta), tantalum nitride (Ta₂N), cobalt (Co) and ruthenium (Ru). The conductive layer **126** may be deposited using various depositing techniques, such as CVD.

[0028] FIG. 1G illustrates the formation of a dielectric layer **130** as well as cut features **132a**, **132b**. The dielectric layer **130**, as well as the cut features **132a**, **132b**, includes at least one of silicon nitride (SiN), silicon oxynitride (SiON), silicon carbon (SiC) and silicon oxycarbide (SiCN). According to the present example, the dielectric layer **130** is deposited over the conductive layer **126**. The dielectric layer **130** may also be referred to as a self-aligned capping layer. After the dielectric layer is deposited, it may be patterned. The patterning process may involve depositing a hard mask layer and a photoresist layer. The photoresist may then be exposed to a light source through a photomask. The photoresist may then be developed to leave a pattern in the photoresist. That pattern may then be transferred to the hard mask. An etching process, such as a dry etching process, may then be applied to the dielectric layer **130** through the hard mask. That etching

process may form trenches that extend all the way to the dummy fin structure. In other words, the etching process may remove both portions of the dielectric layer **130** and portions of the conductive layer **126** to expose top surfaces of some of the dummy fin structures. Thus, the etching process “cuts” the conductive layer.

[0029] After the etching process is performed, cut features **132a**, **132b** are formed within the trench left by the etching process. The cut features **132a**, **132b** may include a dielectric material and in some examples may be the same material as the dielectric layer **130**. In some examples, however, the cut features **132a**, **132b** may include a different dielectric feature.

[0030] The cut features **132a**, **132b** electrically isolate different metal gate segments **124**. For example, cut feature **132a** electrically isolates metal gate segments **124b** and **124c**. Similarly, cut feature **132b** electrically isolates metal gate segments **124c** and **124d**. Because there is no cut feature above dummy fin structure **114a**, the metal gate segments **124a** and **124b** remain electrically connected through the conductive layer **126**.

[0031] In the present example, the width of cut feature **132a** is less than the width of dummy fin structure **114b** to which it connects. However, the width of cut feature **132b** is greater than the width of dummy fin structure **114c** to which it connects. In some examples, the cut feature **132b** may extend to the top surface of the metal gate segments **124c**, **124d** and thus partially cover the top-most portion of side surfaces of the dummy fin structure **114c**.

[0032] FIG. 2 is a diagram showing a top view of the gate cut feature. FIG. 2 thus illustrates the real fin structures **106a**, **106b**, **106c**, **106d** as well as the dummy fin structures **114a**, **114b**, **114c** which extend along a first direction. The conductive layer **126**, which covers the metal gate segments **124**, extends in a second direction that is perpendicular to the first direction. As can be seen, gate cut feature **132a** is positioned over dummy fin structure **114b** and cuts the portion of conductive layer **126** between fin structures **106b** and **106c**. Similarly, gate cut feature **132b** is positioned over dummy fin structure **114c** and cuts the portion of the conductive layer **126** between fin structures **106c** and **106d**.

[0033] FIG. 3 is a flowchart showing an illustrative method for forming a gate cut feature. According to the present example, the method **300** includes a process **302** for forming a plurality of fin structures (e.g., **106a**, **106b**, **106c**, **106d**) extending in a first direction. The fin structures may include nanostructures such as nanowires or nanosheets. Such structures are used in gate-all-around (GAA) transistor devices. The principles described herein may be applied to fin structures that have a gate on three sides, as well as fin structures that are processed to include nanostructures. The example described herein includes nanostructures.

[0034] To form the fin structures, alternating layers of differing semiconductor materials (e.g., **110**, **112**) may be deposited onto a substrate (e.g., **102**). For example, if the substrate is a silicon substrate, then alternating layers of silicon and silicon germanium (SiGe) may be deposited. Then, a hard mask layer **108** may be deposited on top of the alternating semiconductor layers. The hardmask layer **108** may include at least one of silicon oxide (SiO₂), silicon nitride (SiN), silicon carbide (SiC), silicon oxynitride (SiON), silicon oxycarbide (SiOCN), hafnium oxide (HfO₂), aluminum oxide (Al₂O₃), and zirconium oxide (ZrO₂). Other materials are contemplated. A photoresist material may then be used to pattern the hard mask layer. The photoresist may then be exposed to a light source through a photomask. The photoresist may then be developed such that the portions of the photoresist remain while other portions are removed. The pattern within the developed photomask is then transferred to the hard mask layer, which is then transferred to the substrate and alternating semiconductor layers through an etching process.

[0035] After, the fin structures are formed, an STI layer (e.g., **104**) may be deposited, and then etched back to the desired height. The STI layer separates the fin structures from each other. The STI layer may be recessed to a point approximately halfway between a bottom and a top of the fin structure. However, in some examples, the STI layer may be recessed to different heights along the height of the fin structures.

[0036] According to the present example, the method **300** further includes a process **304** for forming a plurality of dummy fin structures (e.g., **114a**, **114b**, **114c**) between the plurality of fin structures. The dummy fin structures may be formed by several processes. Specifically, a first dielectric material (e.g., **116**) may be conformally deposited within the trenches between the real fin structures, which may be surrounded by a cladding (e.g., **105**). This dielectric material may be, for example, silicon nitride (SiN), silicon oxynitride (SiON), or silicon carbide (SiC). Other dielectric materials may be used. The dielectric material may be deposited using an Atomic Layer Deposition (ALD) process.

[0037] After the conformal dielectric layer is formed, a second dielectric layer (e.g., **118**) is deposited on the first dielectric layer. The second dielectric layer may be an oxide layer. The oxide layer may be, for example, silicon oxide, aluminum oxide, or titanium oxide. Other oxides are contemplated as well. In some examples, the oxide layer may be the same type of material used for the STI layer. The oxide layer may be formed, for example, using a Chemical Vapor Deposition (CVD) process. Other processes may be used as well. After the oxide layer **118** is formed, a Chemical Mechanical Polishing (CMP) process may be applied to planarize the top surface of the workpiece.

[0038] An etching process may then be applied to selectively remove the oxide layer while leaving the cladding layer substantially intact. The etching process may be, for example, a dry etching process. In some examples, the etching process is applied so that, after the etching process, the top surface of the oxide layer is about 5-15 nanometers higher than a top surface of the top channel (e.g., **112**) of the fin structures.

[0039] After the oxide layer has been partially etched back, a high-k dielectric layer (e.g., **120**) may be deposited at the top of the dummy fin structures. The high-k dielectric layer may fill the space left by the etching process that etches back the oxide layer. The high-k dielectric layer may be, for example, hafnium oxide (HfO₂), zirconium oxide (ZrO₂), hafnium aluminum oxide (HfAlO_x), hafnium silicon oxide (HfAlO_x), or aluminum oxide (Al₂O₃). The bottom surface of the high-k dielectric layer may be about 5-15 nanometers higher than a top surface of the top channel of the fin structures.

[0040] The method **300** further includes a process **306** for forming a gate structure over the fin structures. The gate structure (e.g., **124**) may be formed by depositing a metal material over the fin structures. In the case that the fin structures include nanostructures such as nanowires or nanosheets, then the material (e.g., **110**) surrounding the channel nanosheets or nanowires may be removed prior to depositing the gate structure. Also before the gate structure is deposited, various layers surrounding the channels **112** may be applied. Such layers may include a high-k gate layer and/or workfunction layers.

[0041] The method **300** further includes a process **308** for recessing the gate structure to expose top surfaces of the dummy fin structures and to separate the gate structure into a plurality of gate structure segments (e.g., **124a**, **124b**, **124c**, **124d**) positioned along a line extending in a second direction orthogonal to the first direction. This etching process may be, for example, a dry etching process. The etching process exposes the top surface of the dummy fin structures.

[0042] The method further includes a process **310** for depositing a conductive layer (e.g., **126**) over the gate structure segments and the dummy fin structures, the conductive layer electrically connecting the gate structure segments. The conductive layer may include, for example, tungsten. In some examples, a thin layer may be deposited before the conductive layer is deposited. The thin layer may be, for example, titanium nitride (TiN). The conductive layer may be deposited using various depositing techniques, such as CVD.

[0043] The method further includes a process **312** for forming a cut feature (e.g., **132a** or **132b**) above at least one of the dummy fin structures to electrically isolate gate structure segments on both sides of the at least one of the dummy fin structures. The cut feature may be formed within a dielectric layer (e.g., **130**) that is deposited on top of the conductive layer. The dielectric layer, as

well as the cut feature, may include, for example, silicon nitride (SiN). The dielectric layer may be deposited over the conductive layer. After the dielectric layer is deposited, it may be patterned. The patterning process may involve depositing a hard mask layer and a photoresist layer. The photoresist may then be exposed to a light source through a photomask. The photoresist may then be developed to leave a pattern in the photoresist. That pattern may then be transferred to the hard mask. An etching process, such as a dry etching process, may then be applied to the dielectric layer through the hard mask. That etching process may form trenches that extend all the way to the dummy fin structure. In other words, the etching process may remove both portions of the dielectric layer and portions of the conductive layer to expose top surfaces of some of the dummy fin structures. Thus, the etching process “cuts” the conductive layer. After the etching process is performed, the cut feature is formed within the trench left by the etching process. The cut feature may include a dielectric material and, in some examples, may be the same material as the dielectric layer. In some examples, the width of the cut feature may be less than the width of dummy fin structure to which it connects. However, in some examples the width of cut feature may be greater than the width of dummy fin structure to which it connects.

[0044] Thus, according to principles described herein, the spacing constraints for cut features and alignment process is substantially improved by changing the depth at which cut features are able to be made. In comparison with other approaches, a combination of the dummy fin structure and cut feature further reduces a cell height, thereby increasing pattern density with respect to circuit design. Using the principles described herein, the overlay constraints for cut features are reduced because the cut feature may be placed anywhere over a dummy fin structure. Furthermore, the circuit can be designed by spacing fin structures closer together, even where gate features are to be cut.

[0045] According to one example, a semiconductor structure includes a plurality of fin structures extending along a first direction, a plurality of gate structure segments positioned along a line extending in a second direction, the second direction being orthogonal to the first direction, wherein the gate structure segments are separated by dummy fin structures. The semiconductor structure further includes a conductive layer disposed over both the gate structure segments and the dummy fin structures to electrically connect at least some of the gate structure segments, and a cut feature aligned with one of the dummy fin structures and positioned to electrically isolate gate structure segments on both sides of the one of the dummy fin structures.

[0046] According to one example, a semiconductor structure includes a first gate structure segment and a second gate structure segment extending along a line in a first direction and separated by a first dummy fin structure extending in a second direction orthogonal to the first direction, a third gate structure segment separated from the second gate structure segment by a second dummy fin structure extending along the second direction, a conductive layer connecting the first gate structure segment and the second gate structure segment, and a cut feature positioned above the second dummy fin structure and isolating the second gate structure segment from the third gate structure segment.

[0047] According to one example, a method includes forming a plurality of fin structures extending in a first direction, forming a plurality of dummy fin structures between the plurality of fin structures, forming a gate structure over the fin structures, recessing the gate structure to expose top surfaces of the dummy fin structures and to separate the gate structure into a plurality of gate structure segments positioned along a line extending in a second direction orthogonal to the first direction, depositing a conductive layer over the gate structure segments and the dummy fin structures, the conductive layer electrically connecting the gate structure segments, and forming a cut feature above at least one of the dummy fin structures to electrically isolate gate structure segments on both sides of the at least one of the dummy fin structures.

[0048] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate

that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor structure comprising: a first active region and a second active region; a first gate structure segment and a second gate structure segment over the first active region and the second active region respectively; a dummy structure interposing the first gate structure segment and the second gate structure segment; a conductive layer disposed over both the first gate structure segment, the dummy structure, and the second gate structure segment; and a dielectric cut feature over the dummy structure and defining a sidewall of the conductive layer over the dummy structure.
2. The semiconductor structure of claim 1, wherein the first active region and the second active region are each fin structures and the dummy structure is a fin structure.
3. The semiconductor structure of claim 1, wherein the conductive layer is extending in a second direction in a top view, the first active region is extending in a first direction in the top view, and the first gate structure segment is extending in the second direction in the top view.
4. The semiconductor structure of claim 3, wherein the first direction is different than the second direction.
5. The semiconductor structure of claim 1, wherein the dummy structure is a multi-layer dielectric structure.
6. The semiconductor structure of claim 1, wherein a shallow trench isolation feature is disposed directly below the dummy structure.
7. The semiconductor structure of claim 1, wherein the first active region is a vertical stack of channel layers.
8. A semiconductor device comprising: a first dummy structure disposed over a substrate, the first dummy structure extending in a first direction in a top view; a second dummy structure disposed over the substrate, the second dummy structure extending in the first direction in the top view; a first active region disposed over the substrate and extending in the first direction in the top view and disposed between the first dummy structure and a first side of the second dummy structure; a second active region disposed over the substrate and extending in the first direction in the top view and disposed adjacent a second side of the second dummy structure; a first gate structure engaging the first active region; a second gate structure engaging the second active region; a conductive layer extending in a second direction in the top view extending over the first dummy structure, the second dummy structure, the first active region, and the second active region; and a dielectric feature disposed over the second dummy structure and defining a sidewall of the conductive layer over the second dummy structure in the top view.
9. The semiconductor device of claim 8, wherein the first dummy structure is formed of a plurality of dielectric material layers including at least an oxide layer and a high-k dielectric layer.
10. The semiconductor device of claim 8, wherein a plane extending through a top surface of the first dummy structure is above a plane extending from a top surface of either of the first gate structure and the second gate structure in a cross-sectional view.
11. The semiconductor device of claim 8, wherein the dielectric feature has a rectangular shape in the top view.
12. The semiconductor device of claim 11, wherein the dielectric feature has a rectangular shape in a cross-sectional view.

13. A semiconductor structure comprising: a first active region and a second active region, wherein in a cross-sectional view a space interposes the first and second active regions; a first dielectric structure disposed in the space; a first gate structure segment engaging the first active region and disposed in the space; a second gate structure segment engaging the second active region and disposed in the space, wherein the second gate structure segment has a first sidewall adjacent the first dielectric structure and a second sidewall adjacent a second dielectric structure; and a conductive layer extending from over the first gate structure segment to over the second dielectric structure.

14. The semiconductor structure of claim 13, wherein the first active region is a plurality of vertically stacked nanostructures and the second active region is another plurality of vertically stacked nanostructures in a cross-sectional view.

15. The semiconductor structure of claim 14, wherein the first active region extends in a first direction in a top view and the second active region extends in the first direction in the top view, wherein the space extends in a second direction different from the first direction.

16. The semiconductor structure of claim 13, wherein the conductive layer contiguously extends over the first gate structure segment and the first dielectric structure.

17. The semiconductor structure of claim 13, wherein the conductive layer contiguously extends from over the first dielectric structure to over the second gate structure segment.

18. The semiconductor structure of claim 13, wherein the conductive layer contiguously extends over the second gate structure segment to over the second dielectric structure.

19. The semiconductor structure of claim 18, wherein the conductive layer has a terminal end over the second dielectric structure.

20. The semiconductor structure of claim 13, wherein the conductive layer includes at least one of titanium (Ti), titanium nitride (TiN), tantalum (Ta), tantalum nitride (TaN), cobalt (Co), or ruthenium (Ru).
